Akanksha Chukka

Associate Data Scientist

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Profile Summary

- Data scientist with about 1 year of hands-on experience using **Python**, **SQL**, **PySpark**, and **Microsoft Azure** to turn large operational datasets into practical insights and models.
- Tend to start by sitting with stakeholders and clarifying what decision they are actually trying to make, then shape the **Azure Databricks** and **Azure SQL Database** exploration around that question so findings are concrete and not just interesting charts.
- Usually end up owning the unglamorous data work, wiring **Azure Data Factory** pipelines, cleaning inconsistent VINs, and using **PySpark**, **Pandas**, and plain **SQL** to get reliable, analysis ready data before talking about models.
- Comfortable comparing models and tradeoffs rather than chasing a single metric, using **scikit-learn**, **Random Forest**, **XGBoost**, and **Logistic Regression** to test ideas, tuning in **Azure Machine Learning** or **Databricks**, and then explaining to non technical partners why a slightly less accurate model might still be the right fit for production.

Technical Skills

Python: pandas, NumPy, scikit-learn, statsmodels, Matplotlib, Seaborn, JupyterLab, PySpark, MLflow
Machine Learning: XGBoost, LightGBM, RandomForestClassifier, LogisticRegression, KMeans, GridSearchCV
SQL & Data: Azure SQL Database, T-SQL, PostgreSQL, Azure Data Factory, Azure Blob Storage, Parquet, CSV
Azure ML Stack: Azure Machine Learning, Azure Databricks, Azure Functions, Azure Cognitive Services, Azure DevOps
JavaScript & Frontend: JavaScript (ES6+), TypeScript, React, Node.js, npm
Data Ops & Versioning: Git, GitHub, Azure Repos, Docker

Education

University of West Florida — Pensacola, FL
Master of Science in Data Science — Jan 2024 – Dec 2025

Work Experience

Wipro — Hyderabad, India
Data Scientist — Dec 2022 – Jan 2024

- Used **Azure Databricks** with **PySpark** and **Azure SQL Database** to explore multi-million row vehicle service datasets and surface pattern and anomaly candidates for business review.
- Given noisy dealer service feeds and inconsistent VIN formats, built **Azure Data Factory** pipelines and Python routines that automated cleansing, deduplication, and gap-filling for downstream modeling.
- Investigated underperforming churn and parts demand models, tuned tree-based and gradient boosting hyperparameters, and rebalanced training data to improve practical prediction accuracy in production settings.
- During a joint initiative with operations analysts, prototyped and iterated **Azure Machine Learning** workflows that trained, evaluated, and versioned predictive models with explicit tracking of feature impacts.
- Implemented anomaly detection logic in **Azure Functions** backed by **Azure Blob Storage** datasets, flagging unusual warranty claim patterns and feeding alerts into existing business monitoring channels.

Software Projects

Credit Card Fraud Detection Using Machine Learning

- Developed supervised machine learning models to spot fraudulent credit card transactions using **Python**, **SQL**, and **PySpark**. Had to tweak the models a bit to get that 95% detection rate with fewer false positives.
- Worked on data preprocessing and feature extraction with **Databricks**, tackling issues like imbalanced data using **SMOTE** and undersampling. This helped improve model robustness by about 15% over the baseline.
- Trained and compared models like **Logistic Regression**, **Random Forest**, and **XGBoost**, using **Unity Catalog** for data management. Regular feedback sessions with the team kept us aligned on what was working.

Customer Churn Prediction using Machine Learning

- Built models to predict customer churn with **Python**, **Pandas**, and **scikit-learn**, hitting an accuracy of around 82-86%. Communicating complex data insights to stakeholders was a big part of the process.
- Did a deep dive into exploratory data analysis and feature engineering on customer behavior data, which boosted model performance by about 10-12%. It was all about paying attention to the details.
- Trained models like **Logistic Regression**, **Random Forest**, and **XGBoost**, optimizing hyperparameters through cross-validation. We used **Microsoft Azure** for deployment, which helped streamline our workflow.